



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Topology of Metric Spaces

Open sets, Limit points and Closed sets

FEBRUARY 15, 2026

CONDUCTED BY

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

✉ partho.dhor@bracu.ac.bd | ✉ parthosutradhor@gmail.com

For updates subscribe on  [@ParthoSutraDhor](#)

Open and Closed Spheres

Open Sphere

Let (X, d) be a metric space and let $a \in X$. For any $r > 0$, the set

$$S_r(a) = \{x \in X : d(x, a) < r\}$$

is called an *open sphere* (or *open ball*) of radius r centered at a .

Closed Sphere

Let (X, d) be a metric space and let $a \in X$. For any $r > 0$, the set

$$S_r[a] = \{x \in X : d(x, a) \leq r\}$$

is called a *closed sphere* of radius r centered at a .

Relation Between Open and Closed Spheres

Show that

$$S_r(a) \subseteq S_r[a]$$

for every $a \in X$ and for every $r > 0$.

Neighbourhood of a Point

Neighbourhood of a Point

Let (X, d) be a metric space and let $a \in X$. A subset $N_a \subseteq X$ is called a *neighbourhood* of the point a if there exists an open sphere $S_r(a)$ centered at a such that

$$S_r(a) \subseteq N_a$$

for some $r > 0$.

② Open Sphere Is a Neighbourhood of Its Points

Show that every open sphere is a neighbourhood of each of its points.

Open Set

Open Set

A subset G of a metric space (X, d) is said to be *open* in X (with respect to the metric d) if G is a neighbourhood of each of its points. In other words, G is open if for each $a \in G$ there exists an $r > 0$ such that

$$S_r(a) \subseteq G.$$

❓ Illustrations of Open Sets

Show that each of the following subsets is open. 

1. The empty set $\emptyset$ and the entire space X with any metric are open sets.
2. Every open sphere in a metric space (X, d) is an open set.
3. Let $A = \{z \in \mathbb{C} : 1 < |z| < 2\}$ with the usual metric. Then A is an open set.

4. Let

$$S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$$

with the Euclidean metric. Then S is an open set.

5. Let

$$A = \{(x, y) \in \mathbb{R}^2 : y^2 < x\}$$

with the Euclidean metric. Then A is an open set.

6. Let

$$A = \left\{ f \in C[a, b] : \inf_{x \in [a, b]} f(x) > 0 \right\}.$$

Then A is open in $C[a, b]$ with respect to the supremum metric.

② Singleton Set in $\mathbb{R}$ Is Not Open

Show that on the real line with the usual metric, the singleton set $\{x\}$ is not open.

② Singleton Sets Under Different Metrics

Let $\mathbb{R}$ be the set of real numbers, d the usual metric and d' the discrete metric on the same set $\mathbb{R}$. Show that every singleton set $\{x\}$, $x \in \mathbb{R}$, is open in $(\mathbb{R}, d')$ but not open in $(\mathbb{R}, d)$.

② Discrete Metric Space: Every Set Is Open

Show that every subset of a discrete metric space (X, d) is open.



② A Subset Open in a Subspace

Let (X, d) be a metric space where $X = (0, 2)$ with the usual metric. Show that the subset

$$A = (0, 1)$$

is an open set in X .



Basic Properties of Open Sets

In any metric space (X, d) ,



1. the union of an arbitrary family of open sets is open;
2. the intersection of a finite number of open sets is open.

② Intersection of Infinite Open Sets May Not Be Open

Show that the intersection of an infinite number of open sets in a metric space need not be open.

Limit Point

Limit Point

Let (X, d) be a metric space and let $A \subseteq X$. A point $x \in X$ is called a *limit point* (or *accumulation point*) of the set A if for every $r > 0$,

$$S_r(x) \cap (A \setminus \{x\}) \neq \emptyset.$$

② Illustrations of Limit Points and Derived Sets

Determine the set of limit points (derived set) in each of the following cases.

1. Let $X = \mathbb{R}$ with the usual metric and let

$$A = [0, 1).$$

Every point of A is a limit point of A , and 1 is also a limit point of A although $1 \notin A$. Hence

$$A' = [0, 1].$$

2. Let

$$A = \left\{1, \frac{1}{2}, \frac{1}{3}, \dots\right\}$$

with the usual metric in $\mathbb{R}$. Then 0 is the only limit point of A and $0 \notin A$. Hence

$$A' = \{0\}.$$

3. If (X, d) is a discrete metric space, then the derived set of every subset of X is empty, that is,

$$A' = \emptyset \quad \text{for all } A \subseteq X.$$

4. Every real number is a limit point of the set of rational numbers $\mathbb{Q}$.
Hence

$$\mathbb{Q}' = \mathbb{R}.$$

5. The set of integers $\mathbb{Z}$ has no limit point, that is,

$$\mathbb{Z}' = \emptyset.$$

6. Every finite subset of a metric space has no limit point.

7. Let

$$A = \{z \in \mathbb{C} : 1 < |z| < 2\}$$

with the usual metric. Then

$$A' = \{z \in \mathbb{C} : 1 \leq |z| \leq 2\}.$$

Closed Sets

Closed Set

A subset F of a metric space (X, d) is said to be *closed* if F contains all its limit points.

❓ Illustrations of Closed Sets

Verify the following statements concerning closed sets in metric spaces.

1. The empty set $\emptyset$ and the whole space X are closed sets in every metric space (X, d) .
2. On the real line $\mathbb{R}$ with the usual metric, the set $\mathbb{N}$ of natural numbers is closed.
3. The set $\mathbb{Q}$ of rational numbers is not closed in $\mathbb{R}$ with the usual metric.
4. Every closed interval on the real line is a closed set.
5. Every finite subset of a metric space is closed.
6. Let

$$A = \{f \in C[a, b] : f(a) = 1\}.$$

Then A is closed in $C[a, b]$ with respect to the supremum metric.

7. The set

$$\left\{ \frac{1+i}{n} : n \in \mathbb{N} \right\}$$

is neither open nor closed in the complex plane with respect to the usual metric.

Equivalent Definition of Closed Sets

Let (X, d) be a metric space. A subset $F \subseteq X$ is closed if and only if its complement $X \setminus F$ is open. 

② Closed Sphere Is a Closed Set

Show that every closed sphere in a metric space (X, d) is a closed set.

Basic Properties of Closed Sets

In any metric space (X, d) ,



1. the intersection of an arbitrary family of closed sets is closed;
2. the union of a finite number of closed sets is closed.

② Union of Infinite Closed Sets May Not Be Closed

Show that the union of an infinite number of closed sets in a metric space need not be closed.

Closure of a Set

Closure

Let A be a subset of a metric space (X, d) . The *closure* of A , denoted by $\overline{A}$ is the smallest closed set containing A , that is,

$$\overline{A} = \bigcap \{F : F \text{ is closed and } A \subseteq F\};$$

Equivalent Definition of Closure

Let A be a subset of a metric space (X, d) . Then

$$\overline{A} = A \cup A'.$$



❓ Illustrations of Closure

Determine the closure of each of the following subsets.

1. Let $X = \mathbb{R}$ with the usual metric and let

$$A = (0, 1) \cup (1, 2).$$

Then

$$\overline{A} = [0, 2].$$

2. Let $X = \mathbb{R}$ with the usual metric and let

$$A = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}.$$

Then

$$\overline{A} = A \cup \{0\}.$$

3. If (X, d) is a discrete metric space and $A \subseteq X$, then

$$\overline{A} = A.$$

4. Let $X = \mathbb{N}$ with the usual metric and let $A = \{1, 2, 3\}$. Then

$$\overline{A} = A.$$

💡 Properties of Closure

Let (X, d) be a metric space and let A, B be any two subsets of X .
Then:

1. A is closed if and only if $A = \overline{A}$;
2. $A \subseteq B$ implies $\overline{A} \subseteq \overline{B}$;
3. $\overline{A \cup B} = \overline{A} \cup \overline{B}$;
4. $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$.
5. $\overline{\overline{A}} = \overline{A}$.
6. $\overline{A} = (\text{ext } A)^c$.
7. $\overline{A} = \text{int}(A^c)^c$.

Thank You!

We'd love your questions and feedback.

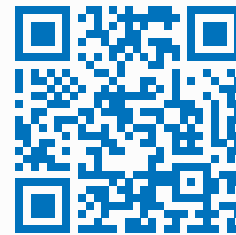
Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

✉ partho.dhor@bracu.ac.bd | ✉ parthosutradhor@gmail.com

 **@ParthoSutraDhor**

(Lectures, walkthroughs, and course updates)



Scan for the channel

References

- [1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.
- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.